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104. On the Theory of Permutants (*Postscript, concluded; book p. 26*)

Postscript. I wish to explain as accurately as I am able, the extent of my obligations to Mr Sylvester in respect of the subject of the present memoir. The term permutant is due to him—intermutant and commutant are merely terms framed between us in analogy with permutant, and the names date from the present year (1851). The theory of commutants is given in my memoir in the Cambridge Philosophical Transactions, [12], and is presupposed in the memoir on Linear Transformations, [13, 14].

It will appear by the last-mentioned memoir that it was by representing the coefficients of a biquadratic function by $a = 1111$, $b = 1112 = 1121 = \&c.$, $c = 1122 = \&c.$, $d = 1222 = \&c.$, $e = 2222$, and forming the commutant

$$\begin{pmatrix} 1111 \\ 2222 \end{pmatrix}$$

that I was led to the function $ae - 4bd + 3c^2$.

The function $ace + 2bcd - ad^2 - b^2e - c^3$ is mentioned in the memoir on Linear Transformations, as brought into notice by Mr Boole. From the particular mode in which the coefficients $a, b, \dots$ were represented by symbols such as 1111, &c., I did not perceive that the last-mentioned function could be expressed in the commutant notation.

The notion of a permutant, in its most general sense, is explained by me in my memoir, “Sur les determinants gauches,” Crelle, t. xxxvii. pp. 93—96, [69]; see the paragraph (p. 94) commencing “On obtient ces fonctions, &c.” and which should run as follows: “On obtient ces fonctions (dont je reprends ici la théorie) par les propriétés générales d’un déterminant défini comme suit. En exprimant &c.,” the sentence as printed being “défini. Car en exprimant &c.,” which confuses the sense. [The paragraph is printed correctly 69, p. 411.]

Some time in the present year (1851) Mr Sylvester, in conversation, made to me the very important remark, that as one of a class the above-mentioned function,

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

would be expressed in the commutant notation, viz. by considering

$$00 = a, \quad 01 = 10 = b, \quad 02 = 11 = 20 = c, \quad 12 = 21 = d, \quad 22 = e;$$

and the subject being thereby recalled to my notice, I found shortly afterwards the expression for the function

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd$$

(which cannot be expressed as a commutant) in the form of an intermutant, and I was thence led to see the identity, so to say, of the theory of hyperdeterminants, as given in the memoir on Linear Transformations, with the present theory of intermutants.

It is understood between Mr Sylvester and myself, that the publication of the present memoir is not to affect Mr Sylvester’s right to claim the origination, and to be considered as the first publisher of such part as may belong to him of the theory here sketched out.

105. Correction of the Postscript to the Paper on Permutants.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 97—98.]

Mr Sylvester has represented to me that the paragraph relating to his communications conveys an erroneous idea of the nature, purport, and extent of such communications; I have, in fact, in the paragraph in question, singled out what immediately suggested to me the expression of the function

$$6abcd + 3b^2c^2 - 4ac^3 - 4b^3d - a^2d^2$$

as a partial commutant or intermutant, but I agree that a fuller reference ought to have been made to Mr Sylvester's results, and that the paragraph in question would more properly have stood as follows:

“Under these circumstances Mr Sylvester communicated to me a series of formal statements, not only oral but in writing, to the effect that he had discovered a permutation method of obtaining as many invariants—viz. commutative invariants—by direct inspection from a function of any degree of any number of letters as the index of the degree contains even factors; one of these commutative invariants being in fact the function

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

expressible, according to Mr Sylvester's notation of my memoir in the *Camb. Phil. Trans.*, supposing $00 = a$, $01 = 10 = b$, $02 = 11 = 20 = c$, $12 = 21 = d$, $22 = e$, by

$$\begin{pmatrix} 00 & 11 & 22 \\ a & b & c \\ b & c & d \\ c & d & e \end{pmatrix},$$

and so on.”

Mr Sylvester and I shall, I have no doubt, be able to agree to a joint statement of any further correction or explanation which may be required.

106. On the Singularities of Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 166—171.]

In the following paper, for symmetry of nomenclature and in order to avoid ambiguities, I shall, with reference to plane curves and in various phrases and compound words, use the term “node” as synonymous with double point, and the term “spinode” as synonymous with cusp.

I shall, besides, have occasion to consider the several singularities which I call the “flecnode,” the “oscnode,” the “fleflecnode,” and the “tacnode:” the flecnode is a double point which is a point of inflexion on one of the branches through it; the oscnode is a double point which is a point of osculation on one of the branches through it; the fleflecnode is a double point which is a point of inflexion on each of the branches through it; and the tacnode is a double point where two branches touch.

And it may be proper to remark here, that a tacnode may be considered as a point resulting from the coincidence and amalgamation of two double points (and therefore equivalent to twelve points of inflexion) or, in a different point of view, as a point uniting the characters of a spinode and a flecnode.

I wish to call to mind here the definition of conjugate tangent lines of a surface, viz. that a tangent to the curve of contact of the surface with any circumscribed developable and the corresponding generating line of the developable, are conjugate tangents of the surface.

Suppose, now, that an absolutely arbitrary surface of any order be intersected by a plane: the curve of intersection has not in general any singularities other than such as occur in a perfectly arbitrary curve of the same order; but as a plane can be made to satisfy one, two, or three conditions, the curve may be made to acquire singularities which do not occur in such absolutely arbitrary curve.

Let a single condition only be imposed on the plane. We may suppose that the curve of intersection has a node; the plane is then a tangent plane and the node is the point of contact—of course any point on the surface may be taken for the node. We may if we please use the term “nodes of a surface,” “node-planes of a surface,” as synonymous with the points and tangent planes of a surface. And it will be convenient also to use the word node-tangents to denote the tangents to the curve of intersection at the node; it may be noticed here that the node-tangents are conjugate tangents of the surface.

Next let two conditions be imposed upon the plane: there are three distinct cases to be considered.

First, the curve of intersection may have a flecnode. The plane (which is of course still a tangent plane at the flecnode) may be termed a flecnode-plane; the flecnodes are singular points on the surface lying on a curve which may be termed the “flecnode-curve,”¹ and the flecnode-planes give rise to a developable which may be termed the flecnode-develope. The “flecnode-tangents” are the tangents to the curve of intersection at the flecnode; the tangent to the inflected branch may be termed the “singular flecnode-tangent,” and the tangent to the other branch the “ordinary flecnode-tangent.”

Secondly, the curve of intersection may have a spinode. The plane (which is of course still a tangent plane at the spinode) may be termed a spinode-plane; the spinodes are singular points on the surface lying on a curve which may be termed the “spinode-curve.”² And the

¹The flecnode-curve, defined as the locus of the points through which can be drawn a line meeting the surface in four consecutive points, was, so far as I am aware, first noticed in Mr Salmon's paper “On the Triple Tangent Planes of a Surface of the Third Order” (Journal, t. iv. [1849], pp. 252—260), where Mr Salmon, among other things, shows that the order of the surface being n , the curve in question is the intersection of the surface with a surface of the order $11n - 24$.

²The notion of a spinode, considered as the point where the indicatrix is a parabola (on which account the spinode has been termed a parabolic point) may be found in Dupin's *Developpements de Geometrie*: the most important step in the theory of these points is contained in Hesse's memoir “Ueber die Wendepuncte der Curven dritter Ordnung” (Crelle, t. xxviii. [1848], pp. 97—107), where it is shown that the spinode-curve is the curve of

spinode-planes give rise to a developable which may be termed the “spinode-developpe.” Also the “spinode-tangent” is the tangent to the curve of intersection at the spinode.

Thirdly, the curve of intersection may have two nodes, or what may be termed a “node-couple.” The plane (which is a tangent plane at each of the nodes and therefore a double tangent plane) may be also termed a “node-couple-plane.” The node-couples are pairs of singular points on the surface lying in a curve which may be termed the “node-couple-curve,” and the node-couple-planes give rise to a developable which may be termed the “node-couple-developpe.” The tangents to the curve of intersection at the two nodes of a node-couple might, if the term were required, be termed the “node-couple-tangents.” Also one of the nodes of a node-couple may be termed a “node-with-node,” and the tangents to the curve of intersection at such point will be the “node-with-node-tangents.”

It is hardly necessary to remark that the flecnode-curve is not the edge of regression of the flecnode-developpe, and the like remark applies *mutatis mutandis* to the spinode-curve and the node-couple curve.

Finally, let three conditions be imposed upon the plane: there are six distinct cases to be considered, in each of which we have no longer curves and developes, but only singular points and singular tangent planes determinate in number.

First, the curve of intersection may have an oscnode. The plane (which continues a tangent plane at the oscnode) is an “oscnode-plane.” The “oscnode-tangents” are the tangents to the curve of intersection at the oscnode; the tangent to the osculating branch is the “singular oscnode-tangent;” and the tangent to the other branch the “ordinary oscnode-tangent.”

Secondly, the curve of intersection may have a fleflecnode. The plane (which continues a tangent plane at the fleflecnode) is a “fleflecnode-plane.” The “fleflecnode-tangents” are the tangents to the curve of intersection at the fleflecnode.

Thirdly, the curve of intersection may have a tacnode. The plane (which continues a tangent plane at the tacnode) is a “tacnode-plane.” The “tacnode-tangent” is the tangent to the curve of intersection at the tacnode.

Fourthly, the curve of intersection may have a node and a flecnode, or what may be termed a node-and-flecnode. The plane (which is a tangent plane at the node and also at the flecnode, where it is obviously a flecnode-plane) is a “node-and-flecnode-plane.” The “node-and-flecnode-tangents,” if the term were required, would be the tangents to the curve of intersection at the node and at the flecnode of the node-and-flecnode. The node of the node-and-flecnode may be distinguished as the node-with-flecnode, and the flecnode as the flecnode-with-node, and we have thus the terms “node-with-flecnode-tangents,” “flecnode-with-node-tangents,” “singular flecnode-with-node-tangent,” and “ordinary flecnode-with-node-tangent.”

Fifthly, the curve of intersection may have a node and also a spinode, or what may be termed a “node-and-spinode.” The plane (which is a tangent plane at the node, and is also a tangent plane at the spinode, where it is obviously a spinode-plane) is a “node-and-spinode-plane.” The node-and-spinode-tangents, if the term were required, would be the tangents at the node and the tangent at the spinode of the node-and-spinode to the curve of intersection. The node of the node-and-spinode may be distinguished as the “node-with-spinode,” and the spinode as the “spinode-with-node,” and we have thus the terms “node-with-spinode-tangent,” “spinode-with-node-tangent.”

Sixthly, the curve of intersection may have three nodes, or what may be termed a “node-triplet.” The plane (which is a triple tangent plane touching the surface at each of the nodes) is a “node-triplet-plane.” The “node-triplet-tangents,” if the term were required, would be the tangents to the curve of intersection at the nodes of the node-triplet. Each node of the node-triplet may be distinguished as a “node-with-node-couple,” and the tangents to the curve of

intersection of the surface supposed as before of the order n , with a certain surface of the order $4(n-2)$. See also Mr Salmon’s memoir “On the Condition that a Plane should touch a surface along a Curve Line” (Journal, t. iii. [1848], pp. 44—46).

intersection at such nodes are “node-with-node-couple-tangents.” The terms “node-couple-with-node,” “node-couple-with-node-tangent,” might be made use of if necessary.

It should be remarked that the oscnodes lie on the flecnode-curve, as do also the flefnodes; these latter points are real double points of the flecnode-curve. The tacnodes are points of intersection and (what will appear in the sequel) points of contact of the flecnode-curve, the spinode-curve, and the node-couple-curve. The spinode-with-nodes are points of intersection of the spinode-curve and node-couple-curve, and the flecnode-with-nodes are points of intersection of the flecnode-curve and node-couple-curve; the node-with-node-couples are real double points (entering in triplets) of the node-couple-curve.

Consider for a moment an arbitrary curve on the surface; the locus of the node-tangents at the different points of this curve is in general a skew surface, which may however, in cases to be presently considered, degenerate in different ways.

Reverting now to the flecnode-curve, it may be shown that the singular flecnode-tangent coincides with the tangent of the flecnode-curve. For consider on a surface two consecutive points such that the line joining them meets the surface in two points consecutive to the first-mentioned two points. The line meets the surface in four consecutive points, it is therefore a singular flecnode-tangent; each of the first-mentioned two points must be on the flecnode-curve, or the singular flecnode-tangent touches the flecnode-curve. The two flecnode-tangents are by a preceding observation conjugate tangents. It follows that the skew surface, locus of the flecnode-tangents, breaks up into two surfaces, each of which is a developable, viz. the locus of the singular flecnode-tangents is the developable having the flecnode-curve for its edge of regression, and the locus of the ordinary flecnode-tangents is the flecnode-developable. Of course at the tacnode, the tacnode-tangent touches the flecnode-curve.

Passing next to the spinode-curve, the spinode-plane and the tangent-plane at a consecutive point along the spinode-tangent are identical, or their line of intersection is indeterminate.³ The spinode-tangent is therefore the conjugate tangent to any other tangent line at the spinode, and therefore to the tangent to the spinode-curve. It follows that the surface locus of the spinode-tangents degenerates into a developable surface twice repeated, viz. the spinode-developable.

Consider the tacnode as two coincident nodes; each of these nodes, by virtue of its constituting, in conjunction with the other, a tacnode, is on the spinode-curve; or, in other words, the tacnode-tangent touches the spinode-curve, and the same reasoning proves that it touches the node-couple-curve. It has already been seen that the tacnode-tangent touches the flecnode-curve; consequently the tacnode is a point, not of simple intersection only, but of contact, of the flecnode-curve, the spinode-curve, and the node-couple-curve.

In virtue of the principle of the spinode-plane being identical with the tangent plane at a consecutive point along the spinode tangent, it appears that the tacnode-plane is a stationary plane, as well of the flecnode-developable as of the spinode-developable, and it would at first sight appear that it must be also a stationary tangent plane of the node-couple-developable. But this is not so; the node-with-node-planes envelope, not the node-couple-developable, but the node-couple-developable twice repeated: the tacnode-plane is in a sense a stationary plane on such duplicate developable, but not in any manner on the single developable. The tacnode-plane is an ordinary tangent plane of the node-couple-developable.

Consider now a spinode-with-node, which we have seen is a point of intersection of the spinode-curve and node-couple-curve. The tangent plane at a consecutive point along the spinode-with-node-tangent, is identical with the spinode-with-node-plane; the curve of intersection of the tangent plane at such consecutive point has therefore a node at the node-with-spinode, or the tangent plane in question is a node-couple-plane, and the point of contact is a point on the node-couple-curve. Consequently the spinode-with-node-tangent touches the node-couple-curve, and thence also the spinode-with-node-plane is a stationary tangent plane of the

³It must not be inferred that the tangent plane at such consecutive point is a spinode-plane; this is obviously not the case.

node-couple-developpe.

It should be remarked that no circumscribed developable can have a stationary tangent plane except the tangent planes at the points where the curve of contact meets the spinode-curve, and any one of these planes is only a stationary plane when the curve of contact touches the spinode-tangent; and that the node-couple-curve and the flecnode-curve do not intersect the spinode-curve except in the points which have been discussed.

Recapitulating, the node-couple-curve and the spinode-curve touch at the tacnodes, and intersect at the spinode-with-nodes: moreover, the tacnode-planes are stationary planes of the spinode-developpe, and the spinode-with-node-planes are stationary planes of the node-couple-developpe. Besides this, the two curves are touched at the tacnodes by the flecnode-curve, and the tacnode-planes are stationary planes of the flecnode-developpe.

107. On the Theory of Skew Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 171—173.]

A SURFACE of the n^{th} order is a surface which is met by an indeterminate line in n points. It follows immediately that a surface of the n^{th} order is met by an indeterminate plane in a curve of the n^{th} order.

Consider a skew surface or the surface generated by a singly infinite series of lines, and let the surface be of the n^{th} order. Any plane through a generating line meets the surface in the line itself and in a curve of the $(n-1)^{\text{th}}$ order. The generating line meets this curve in $(n-1)$ points. Of these points one, viz. that adjacent to the intersection of the plane with the consecutive generating line, is a unique point; the other $(n-2)$ points form a system. Each of the $(n-1)$ points are *sub modo* points of contact of the plane with the surface, but the proper point of contact is the unique point adjacent to the intersection of the plane with the consecutive generating line. Thus every plane through a generating line is an ordinary tangent plane, the point of contact being a point on the generating line.

It is not necessary for the present purpose, but I may stop for a moment to refer to the known theorems that the anharmonic ratio of any four tangent planes through the same generating line is equal to the anharmonic ratio of their points of contact, and that the locus of the normals to the surface along a generating line is a hyperbolic paraboloid.

Returning to the $(n-2)$ points in which, together with the point of contact, a generating line meets the curve of intersection of the surface and a plane through the generating line, these are fixed points independent of the particular plane, and are the points in which the generating line is intersected by other generating lines.

There is therefore on the surface a double curve intersected in $(n-2)$ points by each generating line of the surface—a property which, though insufficient to determine the order of this double curve, shows that the order cannot be less than $(n-2)$.

(Thus for $n = 4$, the above reasoning shows that the double-curve must be at least of the second order: assuming for a moment that it is in any case precisely of this order, it obviously cannot be a plane curve, and must therefore be two non-intersecting lines. This suggests at any rate the existence of a class of skew surfaces of the fourth order generated by a line which always passes through two fixed lines and by some other condition not yet ascertained; and it would appear that surfaces of the second order constitute a degenerate species belonging to the class in question.)

In particular cases a generating line will be intersected by the consecutive generating line. Such a generating line touches the double curve.

Consider now a point not on the surface; the planes determined by this point and the generating lines of the surface are the tangent planes through the point; the intersections of consecutive tangent planes are the tangent lines through the point; and the cone generated by these tangent lines or enveloped by the tangent planes is the tangent cone corresponding to the point. This cone is of the n^{th} class. For considering a line through the point, this line meets the surface in n points, i.e. it meets n generating lines of the surface; and the planes through the line and these n generating lines, are of course tangent planes to the cone: that is, n tangent planes can be drawn to the cone through a given line passing through the vertex.

The cone has not in general any lines of inflexion, or, what is the same thing, stationary tangent planes. For a stationary tangent plane would imply the intersection of two consecutive generating lines of the surface. And since the number of generating lines intersected by a consecutive generating line, and therefore the number of planes through two consecutive generating lines, is finite, no such plane passes through an indeterminate point.

The tangent cone will have in general a certain number of double tangent planes; let this number be χ . We have therefore a cone of the class n , number of double tangent planes χ , number of stationary tangent planes 0. Hence, if m be the order of the cone, α the number of

its double lines, and β the number of its cuspidal or stationary lines,

$$\begin{aligned} m &= n(n-1) - 2\chi, \\ \beta &= 3n(n-2) - 6\chi, \\ \alpha &= \frac{1}{2}n(n-2)(n^2-9) - 2\chi(n^2-n-6) + 2\chi(\chi-1). \end{aligned}$$

This is the proper tangent cone, but the cone through the double curve is *sub modo* a tangent cone, and enters as a square factor into the equation of the general tangent cone of the order $n(n-1)$. Hence, if X be the order of the double curve, and therefore of the cone through this curve,

$$m + 2X = n(n-1),$$

and therefore $X = \chi$; that is, the number of double tangent planes to the tangent cone is equal to the order of the double curve.

It does not appear that there is anything to determine χ ; and if this is so, skew surfaces of the n^{th} order may be considered as forming different families according to the order of the double curve upon them. To complete the theory, it should be added that a plane intersects the surface in a curve of the n^{th} order having χ double points but no cusps.

108. On certain Multiple Integrals connected with the Theory of Attractions.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174—178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," Journal, t. ii. [1847], pp. 219—223, [44], the equation

$$\int \frac{d\xi d\eta \dots}{[(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots + v^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1}(e - s)^q ds}{(1 + s)^{\frac{1}{2}n+q}},$$

where n is the number of variables of the multiple integral, and the condition of the integration is

$$(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots < f^2;$$

also where

$$\frac{(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots}{f^2} - \frac{v^2}{ef^2} = 1,$$

and e is the positive root of

$$\xi^2 + \eta^2 + \dots = v^2.$$

Suppose $f = f_1 = \dots = f_n$, and write $(\alpha - \alpha_1)^2 + \dots = k^2$; we obtain

$$\int \frac{d\xi d\eta \dots}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1}(\epsilon - s)^q ds}{(1 + s)^{\frac{1}{2}n+q}},$$

the limiting condition for the multiple integral being

$$(\xi - \alpha_1)^2 + (\eta - \beta_1)^2 + \dots < f_1^2,$$

and the function e , and limit ϵ , being given by

$$e^2 + \xi^2 + \dots = v^2, \quad \epsilon \text{ denoting, as before, the positive root.}$$

Observing that the quantity under the integral sign on the second side vanishes for $s = \epsilon$, there is no difficulty in deducing, by a differentiation with respect to ξ_1 , the formula

$$\int \frac{dS}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q)} \int_0^\epsilon \frac{s^{q-1}(\epsilon - s)^{q-1} ds}{(1 + s)^{\frac{1}{2}n+q-1}},$$

where dS is the element of the surface $(\xi - \alpha_1)^2 + \dots = f_1^2$ and the integration is extended over the entire surface.

A slight change of form is convenient. We have

$$s^{q-1}(\epsilon - s)^{q-1} ds = \frac{1}{s(1 + s)} \cdot \frac{s^q(\epsilon - s)^q}{\epsilon} d\epsilon,$$

if we suppose

$$\epsilon = \frac{\xi_1^2 + \eta_1^2 + \dots - f_1^2}{(\xi - \alpha_1)^2 + (\eta - \beta_1)^2 + \dots}.$$

The formulae then become

$$\int \frac{dS}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q)} \int_0^\epsilon \frac{(\xi_1^2 + \dots - f_1^2)^{q-1} d\epsilon}{\epsilon(1 + \epsilon)[(\xi - \alpha_1)^2 + \dots]^{\frac{1}{2}n-q-1}},$$

in which ϵ is the positive root of the equation

$$\rho^2 e^2 + \lambda e - \mu = 0.$$

I propose to transform these formulae by means of the theory of images; it will be convenient to investigate some preliminary formulae.

Suppose $l^2 = \alpha^2 + \beta^2 + \dots$, $l_1^2 = \alpha_1^2 + \beta_1^2 + \dots$; also consider the new constants $a, b, \dots, a_1, b_1, \dots, u, \mu$, determined by the equations

$$\frac{S}{u} = \frac{\alpha}{a} = \frac{\beta}{b} = \dots = \frac{l^2}{\mu} = \frac{l_1^2}{S},$$

where S is arbitrary.

Then, putting

$$x = \frac{S\xi}{\xi^2 + \eta^2 + \dots}, \quad y = \frac{S\eta}{\xi^2 + \eta^2 + \dots}, \quad \dots$$

it is easy to see that

$$(x - a)^2 + (y - b)^2 + \dots = \frac{S^2}{(\xi^2 + \eta^2 + \dots)^2} \{(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots\},$$

and

$$\frac{a}{l^2} = \frac{\alpha}{l^2 + uS}, \quad \frac{S^2 \alpha}{l^2 + uS} = a, \quad \frac{S^2 \alpha_1}{l_1^2 + \mu S} = a_1.$$

Proceeding to express the single integrals in terms of the new constants, we have in the first place $k^2 = h^2 \cdot l^2$, where

$$h^2 = \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2} = \frac{l^2 l_1^2 + S(\mu l^2 + u l_1^2) + u \mu S^2}{S^2};$$

or if we write

$$\alpha \alpha_1 + \beta \beta_1 + \dots = ll_1 \cos \omega,$$

we have

$$h^2 = l^2 + l_1^2 - 2ll_1 \cos \omega.$$

Hence also $k^{-2} = h^{-2} \cdot l^{-2}$, where

$$\frac{l^2 + uS}{S^2 \alpha} = \frac{1}{a}, \quad \text{whence} \quad \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1 \cos \omega} = \frac{1}{\cos \omega};$$

$$\frac{(k^2 - \lambda^2)(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1} = \frac{1}{\rho^2},$$

where $\rho^2 = l^2 + l_1^2 - 2ll_1 \cos \omega$, that is $\rho^2 = (a - a_1)^2 + (b - b_1)^2 + \dots$; consequently

$$\frac{1}{k^2} \cdot \frac{l^2 + uS}{l_1^2 + \mu S} = \frac{S^2}{\Pi},$$

where Π is given by

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2) e - u\mu,$$

and it is clear that e will be the positive root of

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2) e - u\mu = 0.$$

It may be noticed that, in the particular case of $u = 0$, the roots of this equation are 0, and $\frac{\mu}{\lambda^2 - \mu}$. Consequently if $\rho^2 - f_1^2$ and $l^2 - f^2$ are of opposite signs, we have $e = 0$; but if $\rho^2 - f_1^2$ and $l^2 - f^2$ are of the same sign, $e = \frac{\mu}{\lambda^2 - \mu}$.

In order to transform the double integrals, considering the new variables $x, y, \dots$, I write $x^2 + y^2 + \dots = r^2$ and

$$\xi = \frac{Sx}{x^2 + y^2 + \dots}, \quad \eta = \frac{Sy}{x^2 + y^2 + \dots}, \quad \dots$$

whence also, if $\xi^2 + \eta^2 + \dots = \rho^2$ (which gives $r\rho = S^2$), we have

$$\frac{d\xi}{\xi} = \frac{dx}{x} - \frac{2x dx + 2y dy + \dots}{x^2 + y^2 + \dots}, \quad \&c.;$$

also it is immediately seen that

$$(\xi - \alpha)^2 + \dots + v^2 = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \},$$

$$(\xi - \alpha_1)^2 + \dots - f_1^2 = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{ (x - a_1)^2 + (y - b_1)^2 + \dots - f_1'^2 \};$$

and from the latter equation it follows that the limiting condition for the first integral is

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1'^2$$

(there is no difficulty in seeing that the sign $<$ in the former limiting condition gives rise here to the sign $>$), and that the second integral has to be extended over the surface $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1'^2$. Also if dS' represent the element of this surface, we may obtain

$$d\xi d\eta \dots = \frac{S^n}{r^{2n}} dx dy \dots, \quad dS = \frac{S^{n-1}}{r^{2(n-1)}} dS';$$

and, combining the above formulae, we obtain

$$\begin{aligned} \int \frac{dx dy \dots}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \}^{\frac{1}{2}n-q}} \\ = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1) (l^2 + u^2)^{\frac{1}{2}n-q}} \int_0^\epsilon \frac{s^{q-1} (e - s)^q ds}{s(1 + s)^{\frac{1}{2}n+q}}, \end{aligned}$$

the limiting condition of the multiple integral being

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1'^2;$$

and

$$\begin{aligned} \int \frac{dS'}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q-1} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \}^{\frac{1}{2}n-q}} \\ = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q) (l^2 + u^2)^{\frac{1}{2}n-q} (l_1^2 - f_1'^2)} \int_0^\epsilon \frac{s^{q-1} (e - s)^{q-1} ds}{(1 + s)^{\frac{1}{2}n+q-1}}, \end{aligned}$$

where dS' is the element of the surface $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1'^2$, and the integration extends over the entire surface.

In these formulae, l , l_1 , ρ , Π denote as follows:

$$\begin{aligned} l^2 &= a^2 + b^2 + \dots, \\ l_1^2 &= a_1^2 + b_1^2 + \dots, \\ \rho^2 &= (a - a_1)^2 + (b - b_1)^2 + \dots, \\ \Pi &= \rho^2 e^2 + (\mu - \lambda^2) e - u\mu, \end{aligned}$$

and e is the positive root of the equation $\Pi = 0$.

The only obviously integrable case is that for which in the second formula $q = \frac{1}{2}$; this gives

$$\int \frac{dS'}{(x^2 + y^2 + \dots)^{\frac{1}{2}n} \{(x - a)^2 + (y - b)^2 + \dots + u^2\}^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n)(l^2 + u^2)^{\frac{1}{2}n-1}(l_1^2 - f_1^2)(1 + e)^{\frac{1}{2}n-1}}.$$

In the case of $u = 0$, we have, as before, when $\rho^2 - f_1^2$ and $l^2 - f^2$ are of opposite signs, $e = 0$, and therefore $1 + e = 1$; but when $\rho^2 - f_1^2$ and $l^2 - f^2$ are of the same sign, the value before found for e gives

$$1 + e = \frac{\rho^2}{\rho^2 - f_1^2} \{l_1^2 + (\rho^2 - f_1^2)(l^2 - f^2)\} / l_1^2.$$

Consider the image of the origin with respect to the sphere $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$; the coordinates of this image are

$$\frac{f_1^2 a_1}{l_1^2}, \quad \frac{f_1^2 b_1}{l_1^2}, \quad \dots,$$

and consequently, if h be the distance of this image from the point $(a, b, \dots)$, we have

$$h^2 = \frac{1}{l_1^2} \{l_1^2 l^2 + (\rho^2 - f_1^2)(l_1^2 - f_1^2)\};$$

whence, by a simple reduction, or the values of the integral are: when $\rho^2 - f_1^2$ and $l^2 - f^2$ have opposite signs,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{l_1^{n-1}};$$

when $\rho^2 - f_1^2$ and $l^2 - f^2$ have the same sign,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{h^{n-2}(\rho^2 - f_1^2)/l_1^{n-1}},$$

where h is the distance from the point $(a, b, \dots)$ of the image of the origin with respect to the sphere $(x - a_1)^2 + \dots = f_1^2$.

Considered in reference to the theory of images, the first formula gives the potential of a multiple layer (the density being inversely as the $(n - 1)^{\text{th}}$ power of the distance) spread over the surface of an $(n - 1)$ -fold space, due to an attracting mass (the law of attraction being the inverse $(n - 1)^{\text{th}}$ power of the distance) situated at a given point; and the second formula gives the potential of the attracting mass itself. It is easy to see that the formula agrees with the ordinary theory of electrical images.

Stone Buildings, August 6, 1850.

109. On the Rationalisation of certain Algebraical Equations.

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97—101.]

Suppose

$$x + y = 0, \quad x^2 = a, \quad y^2 = b;$$

then if we multiply the first equation by 1, xy , and reduce by the two others, we have

$$x + y = 0, \quad bx + ay = 0,$$

from which, eliminating x, y ,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0;$$

which is the equation between a and b ; or, considering x, y as quadratic radicals, the rational equation between x, y . So if the original equation be multiplied by x, y , we have

$$a + xy = 0, \quad b + xy = 0;$$

or, eliminating 1, xy ,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between x, y .

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1, yz, zx, xy , reduce and eliminate the quantities x, y, z, xyz ; we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & . \\ . & c & b & 1 \\ c & . & a & 1 \\ b & a & . & 1 \end{vmatrix} = 0;$$

and again, multiply the equation by x, y, z, xyz , reduce and eliminate the quantities 1, yz, zx, xy , the result is

$$\begin{vmatrix} a & b & c & . \\ a & . & 1 & 1 \\ b & 1 & . & 1 \\ c & 1 & 1 & . \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (Journal, vol. II. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1, $yz, zx, xy, xw, yw, zw, xyzw$, reducing and eliminating the quantities in the outside row,

$$(x, y, z, w, yzw, zwx, wxy, xyz),$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & 1 \\ c & . & a & . & . & 1 & . & 1 \\ b & a & . & . & 1 & . & . & 1 \\ d & . & . & a & . & 1 & 1 & . \\ . & d & . & b & 1 & . & 1 & . \\ . & . & d & c & 1 & 1 & . & . \\ . & . & . & . & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by $x, y, z, w, yzw, zwx, wxy, xyz$, reduce and eliminate the quantities in the outside row

$$(1, yz, zx, xy, xw, yw, zw, xyzw),$$

we have the result

$$\begin{vmatrix} a & . & . & 1 & 1 & 1 & . & . \\ b & 1 & . & . & 1 & . & 1 & . \\ c & 1 & 1 & . & . & . & . & 1 \\ d & . & . & . & 1 & 1 & 1 & . \\ . & d & . & . & . & c & b & 1 \\ . & . & d & . & c & . & a & 1 \\ . & . & . & d & b & a & . & 1 \\ . & a & b & c & . & . & . & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \quad x^3 = a, \quad y^3 = b;$$

and multiply first by $1, x^2y, xy^2$; this gives

$$x + y = 0, \quad x^2y + ay = 0, \quad bx + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & . \\ . & a & 1 \\ b & . & 1 \end{vmatrix} = 0;$$

next multiply by x, y, x^2y^2 ; this gives

$$x^2 + xy = 0, \quad xy + y^2 = 0, \quad bx^2 + ay^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & . & 1 \\ . & 1 & 1 \\ b & a & . \end{vmatrix} = 0;$$

and lastly, multiply by x^2, y^2, xy ; this gives

$$a + x^2y = 0, \quad b + xy^2 = 0, \quad x^2y + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} a & 1 & . \\ b & . & 1 \\ . & 1 & 1 \end{vmatrix} = 0;$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x + y + z = 0, \quad x^3 = a, \quad y^3 = b, \quad z^3 = c,$$

multiply first by 1, xyz , $x^2y^2z^2$, x^2z , y^2x , z^2y , x^2y , y^2z , z^2x , reduce and eliminate the quantities in the outside row

$$(x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy);$$

the result is

$$\begin{vmatrix} 1 & 1 & 1 & . & . & . & . & . & . \\ . & . & . & 1 & 1 & 1 & . & . & . \\ . & . & . & . & . & . & a & b & c \\ . & . & a & 1 & . & . & . & . & 1 \\ b & . & . & . & 1 & . & . & . & 1 \\ . & c & . & . & . & 1 & 1 & . & . \\ . & a & . & 1 & . & . & . & . & 1 \\ . & . & b & . & 1 & . & 1 & . & . \\ c & . & . & . & . & 1 & . & 1 & . \end{vmatrix} = 0;$$

next multiply by x , y , z , y^2z^2 , z^2x^2 , x^2y^2 , x^2yz , y^2zx , z^2xy , reduce and eliminate the quantities in the outside row

$$(x^2, y^2, z^2, yz, zx, xy, xy^2z^2, yz^2x^2, zx^2y^2),$$

the result is

$$\begin{vmatrix} 1 & . & . & . & 1 & 1 & . & . & . \\ . & 1 & . & 1 & . & 1 & . & . & . \\ . & . & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & . & . \\ c & . & a & . & . & . & . & . & 1 \\ b & a & . & . & . & . & . & 1 & . \\ . & . & a & . & . & . & . & 1 & 1 \\ . & b & . & . & . & . & 1 & . & 1 \\ c & . & . & . & . & . & 1 & 1 & . \end{vmatrix} = 0;$$

lastly, multiply by x^2 , y^2 , z^2 , yz , zx , xy , xy^2z^2 , yz^2x^2 , zx^2y^2 , reduce and eliminate the quantities in the outside row

$$(1, xyz, x^2y^2z^2, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x),$$

the result is

$$\begin{vmatrix} a & . & . & . & 1 & . & . & 1 & . \\ b & . & . & 1 & . & 1 & . & . & . \\ c & . & 1 & . & . & . & 1 & . & . \\ . & 1 & 1 & . & . & 1 & . & . & . \\ . & 1 & . & 1 & . & . & 1 & . & . \\ . & 1 & . & . & 1 & . & . & . & 1 \\ . & . & 1 & c & . & b & . & . & . \\ . & . & 1 & . & a & . & . & . & c \\ . & . & 1 & . & b & . & a & . & . \end{vmatrix} = 0;$$

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

110. Note on the Transformation of a Trigonometrical Expression.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61—62.]

The differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol Ω , and expressing it in the form

$$\Omega = \begin{vmatrix} 1 & a+x & (a+x)\sqrt{c+x} \\ 1 & a+y & (a+y)\sqrt{c+y} \\ 1 & a+z & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment $\xi = \sqrt{\frac{a-c}{c+x}}$, etc., and we get

$$\Omega = \begin{vmatrix} 1 & (a-c)(1+\xi^2) & (a-c)^{3/2}(\xi+\xi^3) \\ 1 & (a-c)(1+\eta^2) & (a-c)^{3/2}(\eta+\eta^3) \\ 1 & (a-c)(1+\zeta^2) & (a-c)^{3/2}(\zeta+\zeta^3) \end{vmatrix}.$$

Pulling out the common factors and simplifying,

$$\Omega = (a-c)^{5/2} (\xi-\eta)(\eta-\zeta)(\zeta-\xi) (\xi\eta + \eta\zeta + \zeta\xi - 1);$$

or, replacing ξ, η, ζ by their values, we have identically

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = (a-c)^{5/2} \frac{(\xi-\eta)(\eta-\zeta)(\zeta-\xi)(\xi\eta + \eta\zeta + \zeta\xi - 1)}{1};$$

and the equation

$$\xi\eta + \eta\zeta + \zeta\xi - 1 = 0, \quad \text{i.e.} \quad \sqrt{\frac{a-c}{c+x}}\sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+y}}\sqrt{\frac{a-c}{c+z}} + \sqrt{\frac{a-c}{c+z}}\sqrt{\frac{a-c}{c+x}} - 1 = 0,$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

111. On a Theorem of M. Lejeune-Dirichlet's.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163—165.]

The following formula,

$$\sum q^{ax^2+2bxy+cy^2} \cdot \sum q^{a'x'^2+2b'x'y'+c'y'^2} \dots = \frac{1}{S} \Sigma \epsilon \left(\frac{D}{n} \right) \left(\frac{D}{n'} \right) \dots q^{nn' \dots / P},$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows: On the left-hand side $(a, b, c), (a', b', c'), \dots$ are a system of properly primitive forms to the negative determinant $-D$ (i.e. a system of positive forms); x, y are positive or negative integers including zero, such that in the sum $\sum q^{ax^2+2bxy+cy^2}$, $ax^2 + 2bxy + cy^2$ is prime to $2D$, and similarly in the other sums; q is indeterminate and the summations extend to the values first mentioned, of x and y . On the right-hand side we have to consider the form of D , viz. we have $D = PS^2$ or else $D = 2PS^2$, where S^2 is the greatest square factor in D and where P is odd: this obviously defines P , and the values of S, ϵ , which are always ± 1 (or, as I prefer to express it, are always $\pm$) are given as follows, viz.

$D = PS^2,$	$P \equiv 1 \pmod{4},$	$S, \epsilon = +, +,$
$D = PS^2,$	$P \equiv 3 \pmod{4},$	$S, \epsilon = -, +,$
$D = 2PS^2,$	$P \equiv 1 \pmod{4},$	$S, \epsilon = +, -,$
$D = 2PS^2,$	$P \equiv 3 \pmod{4},$	$S, \epsilon = -, -,$

n, n' are any positive numbers prime to $2D$, $\left(\frac{D}{n}\right)$ is Legendre's symbol as generalized by Jacobi, viz. in general if p be a positive or negative prime not a factor of n , then $\left(\frac{n}{p}\right) = +1$ or -1 according as n is or is not a quadratic residue of p (or, what is the same thing, p being positive, $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod{p}$), and for $P = pp'p'' \dots$,

$$\left(\frac{n}{P}\right) = \left(\frac{n}{p}\right) \left(\frac{n}{p'}\right) \left(\frac{n}{p''}\right) \dots$$

and the summation extends to all the values of n, n' of the form above mentioned. In the particular case $D = -1$, it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$$D = -1, \quad \sum q^{x^2+y^2} = 4 \Sigma \left(\frac{-1}{n}\right) q^n,$$

$$\text{or} \quad 2(1+2q+2q^4+2q^9+\dots)(q+q^4+q^9+q^{16}+\dots) = \frac{q}{1-q} - \frac{q^3}{1-q^3} + \frac{q^5}{1-q^5} - \frac{q^7}{1-q^7} + \&c.$$

$$D = -2, \quad \sum q^{x^2+2y^2} = 2 \Sigma \left(\frac{-2}{n}\right) q^n,$$

$$\text{or} \quad (1+2q^2+2q^8+2q^{18}\dots)(q+q^3+q^9+q^{17}\dots) = \frac{q}{1-q^2} + \frac{q^3}{1-q^6} - \frac{q^5}{1-q^{10}} - \&c.$$

an example given in the memoir.

For $D = -3$,

$$\begin{aligned} & (q+q^4+q^9+q^{16}+q^{25}\dots)(1+2q^2+2q^6+2q^{12}\dots) + 2(q^2+q^3+q^6+q^{12}+\dots)(q^4+q^9+q^{16}+q^{25}\dots) \\ &= \frac{q}{1-q^2} - \frac{q+q^2}{1-q^3} + \frac{q^3-q^4}{1-q^6} + \frac{q^7-q^8}{1-q^9} + \&c. \end{aligned}$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

112. Demonstration of a Theorem relating to the Products of Sums of Squares.

[From the Philosophical Magazine, vol. IV. (1852), pp. 515—519.]

MR KIRKMAN, in his paper “On Pluquaternions and Homoid Products of Sums of n Squares” (*Phil. Mag.* vol. xxxiii. [1848] pp. 447—459 and 494—509), quotes from a note of mine the following passage: “The complete test of the possibility of the product of 2^m squares by 2^m squares reducing itself to a sum of 2^m squares is the following—forming the complete systems of triplets for $(2^m - 1)$ things, if eab, ecd, fac, fdb be any four of them, we must have, paying attention to the signs alone,

$$(\pm eab)(\pm ecd) = (\pm fac)(\pm fdb);$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things.” I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2 + a^2 + b^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + \dots) = w''^2 + a''^2 + b''^2 + \dots,$$

where the 2^m quantities $w, a, b, c, \dots$ and the 2^m quantities $w_1, a_1, b_1, c_1, \dots$ are given quantities in terms of which the 2^m quantities $w'', a'', b'', c'', \dots$ have to be determined.

Without attaching any meaning whatever to the symbols $a_0, b_0, c_0, \dots$ I write down the expressions

$$ww_1 + aa_1 + bb_1 + cc_1 + \dots, \quad w_1a_0 + a_1b_0 + b_1c_0 + \dots,$$

and I multiply as if $a_0, b_0, c_0, \dots$ really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols a_0, b_0 combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} & ww_1 + (aw_1 + a_1w)a_0 + (bw_1 + b_1w)b_0 + \dots \\ & + aa_1a_0^2 + bb_1b_0^2 + \dots \\ & + ab_1a_0b_0 + a_1b_0a_0 + \dots \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0b_0c_0 = +,$$

means that in the expression of the quasi-product

$$a_0b_0, b_0c_0, c_0a_0, c_0b_0, a_0c_0, b_0a_0$$

are to be replaced by

$$c_0, a_0, b_0, -a_0, -b_0, -c_0;$$

and that a quasi-equation, such as $a_0b_0c_0 = -$, means that in the expression of the quasi-product

$$a_0b_0, b_0c_0, c_0a_0, b_0a_0, c_0b_0, a_0c_0$$

are to be replaced by

$$-c_0, -a_0, -b_0, +a_0, +b_0, +c_0.$$

It is in the first place clear that the quasi-equation $a_0b_0c_0 = +$ may be written in any one of the six forms

$$a_0b_0c_0 = b_0c_0a_0 = c_0a_0b_0 = -a_0c_0b_0 = -c_0b_0a_0 = -b_0a_0c_0,$$

and so for the quasi-equation $a_0b_0c_0 = -$. This being premised, if we form a system of quasi-equations, such as

$$a_0b_0c_0, e_0a_0b_0, \dots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace $a_0^2, b_0^2, \dots$ each by -1 , it is clear that the quasi-product will assume the form

$$w_{\mathcal{H}}w_1 + a_{\mathcal{H}}a_0 + b_{\mathcal{H}}b_0 + c_{\mathcal{H}}c_0 + \dots,$$

$w_{\mathcal{H}}, a_{\mathcal{H}}, b_{\mathcal{H}}, c_{\mathcal{H}}, \dots$ being determinate functions of $w, a, b, c, \dots$; $w_1, a_1, b_1, c_1, \dots$, homogeneous of the first order in the quantities of each set; the value of $w_{\mathcal{H}}$ being obviously in every case

$$w_{\mathcal{H}} = ww_1 - aa_1 - bb_1 - cc_1 - \dots,$$

and $a_{\mathcal{H}}, b_{\mathcal{H}}, c_{\mathcal{H}}, \dots$ containing in every case the terms $aw_1 + a_1w, bw_1 + b_1w, cw_1 + c_1w, \dots$ but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs $\pm$.

[The demonstration continues on book p. 51 et seq., outside the present pp. 26–50 slice.]